SafeMine — Landmine Detection Model
YOLOv8s v2 Report (seed 2)

Generated 2026-10-15 · frozen v2 split · ultralytics 8.4.117

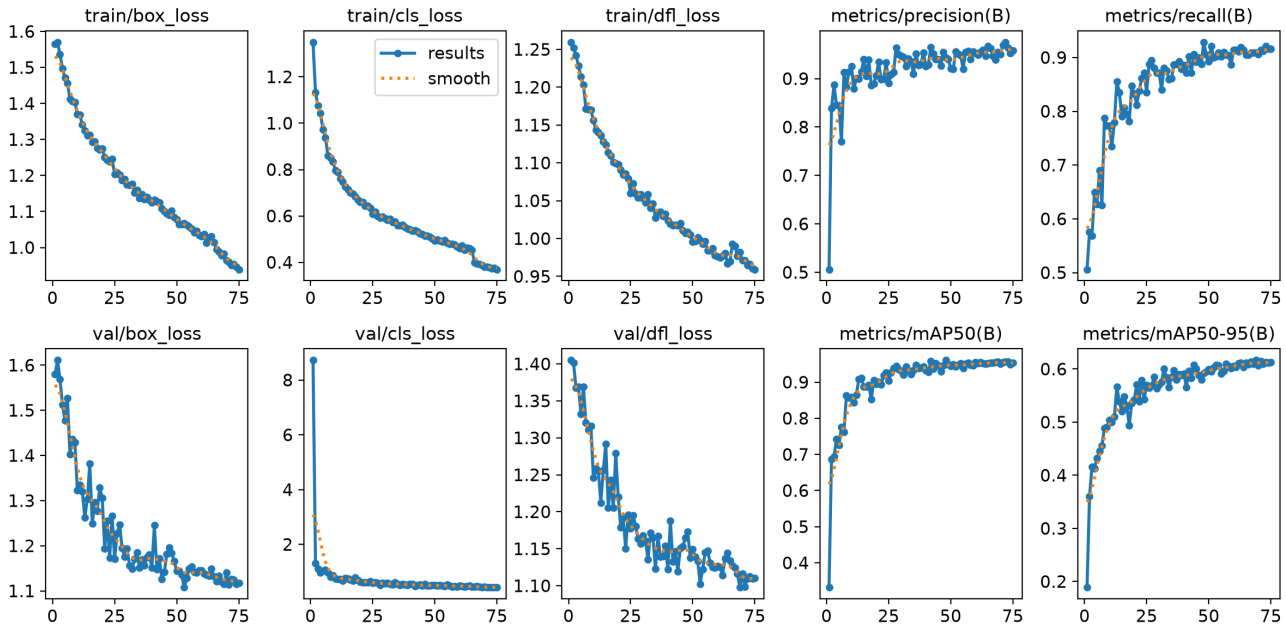
Metric	Validation	Test
Precision (mean)	0.9501	0.9702
Recall (mean)	0.9117	0.9407
mAP@50	0.9574	0.9756
mAP@50-95	0.6172	0.6932
OP conf (val-chosen)	—	0.21
Box P/R @ OP	—	0.7640 / 0.9723
FP/img @ OP	—	0.402
Image-level P/R @ OP	—	0.9945 / 0.9837

Training Configuration

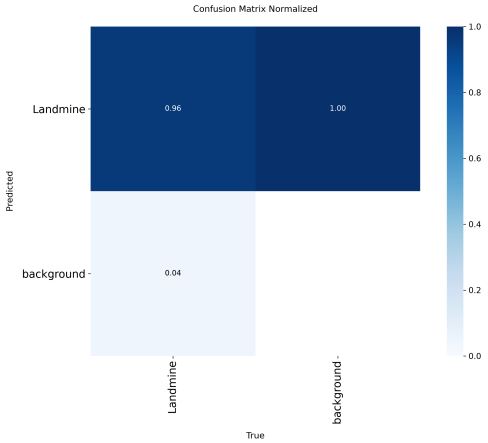
Param	Value
model	yolov8s.pt
epochs	75
imgsz	640
batch	8
seed	2
save_period	5
lr0	0.01
data	C:\Users\Delta\Downloads\Landmine_project\Datasets\landmine_combined_v2\data.yaml

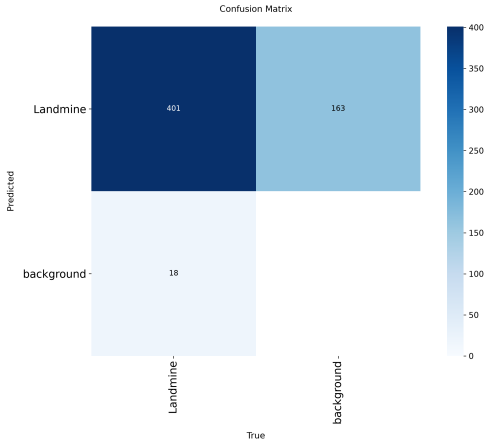
Hardware: NVIDIA GeForce RTX 4070 Laptop GPU (8 GB), FP32, imgsz 640.

Training Curves

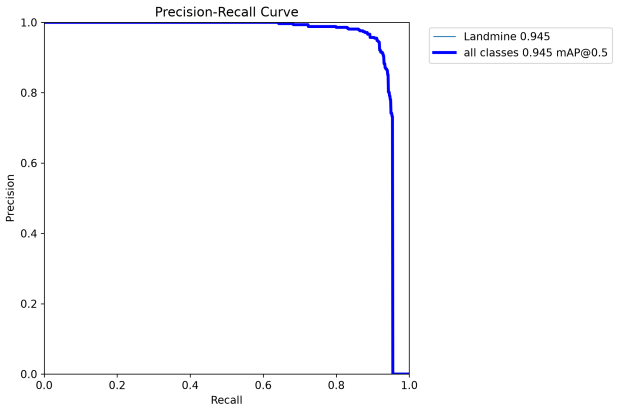


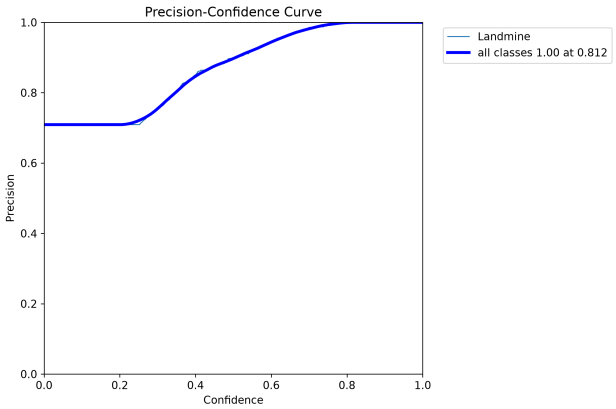
Confusion Matrix (Normalized) — Val

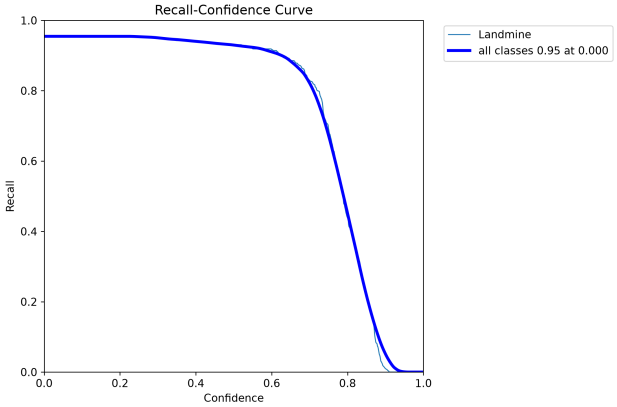


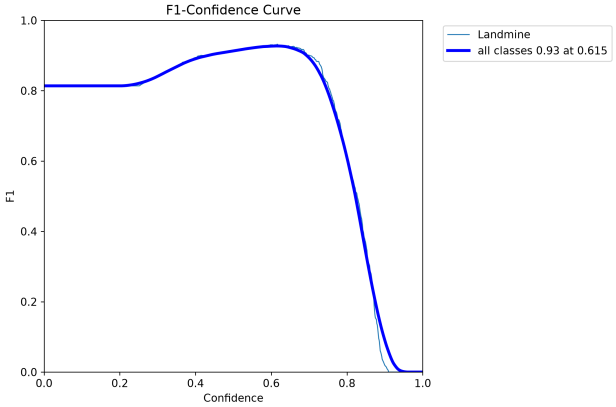


Precision-Recall / P / R / F1 Curves — Val

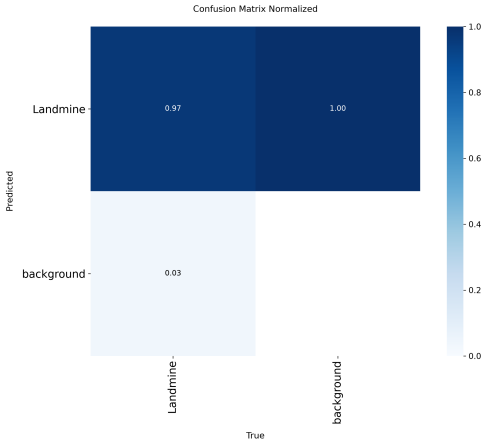


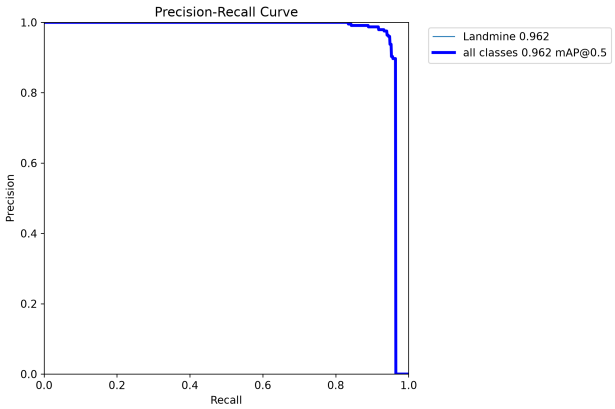




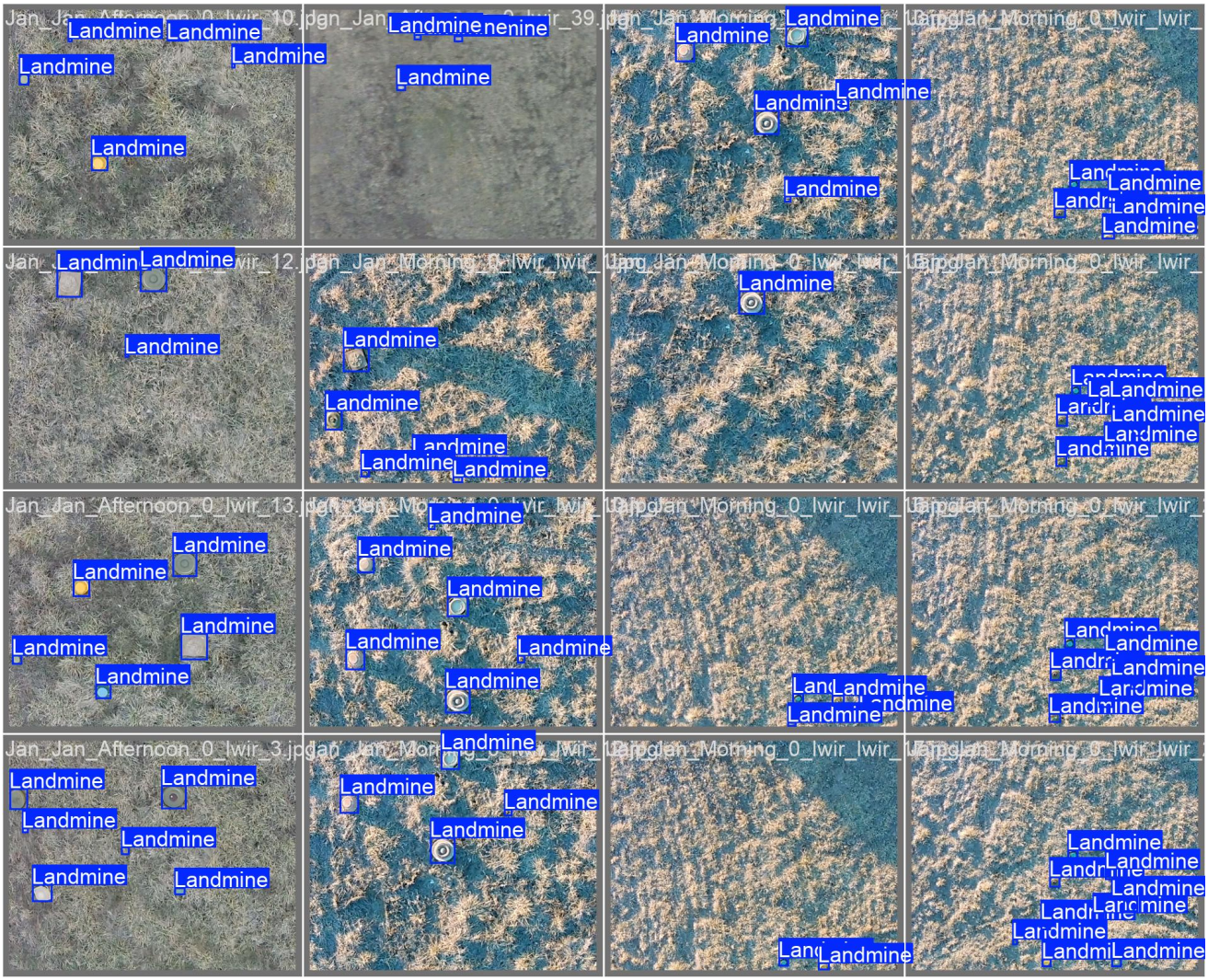


Confusion + Curves — Test (visualization pass @ frozen OP conf)

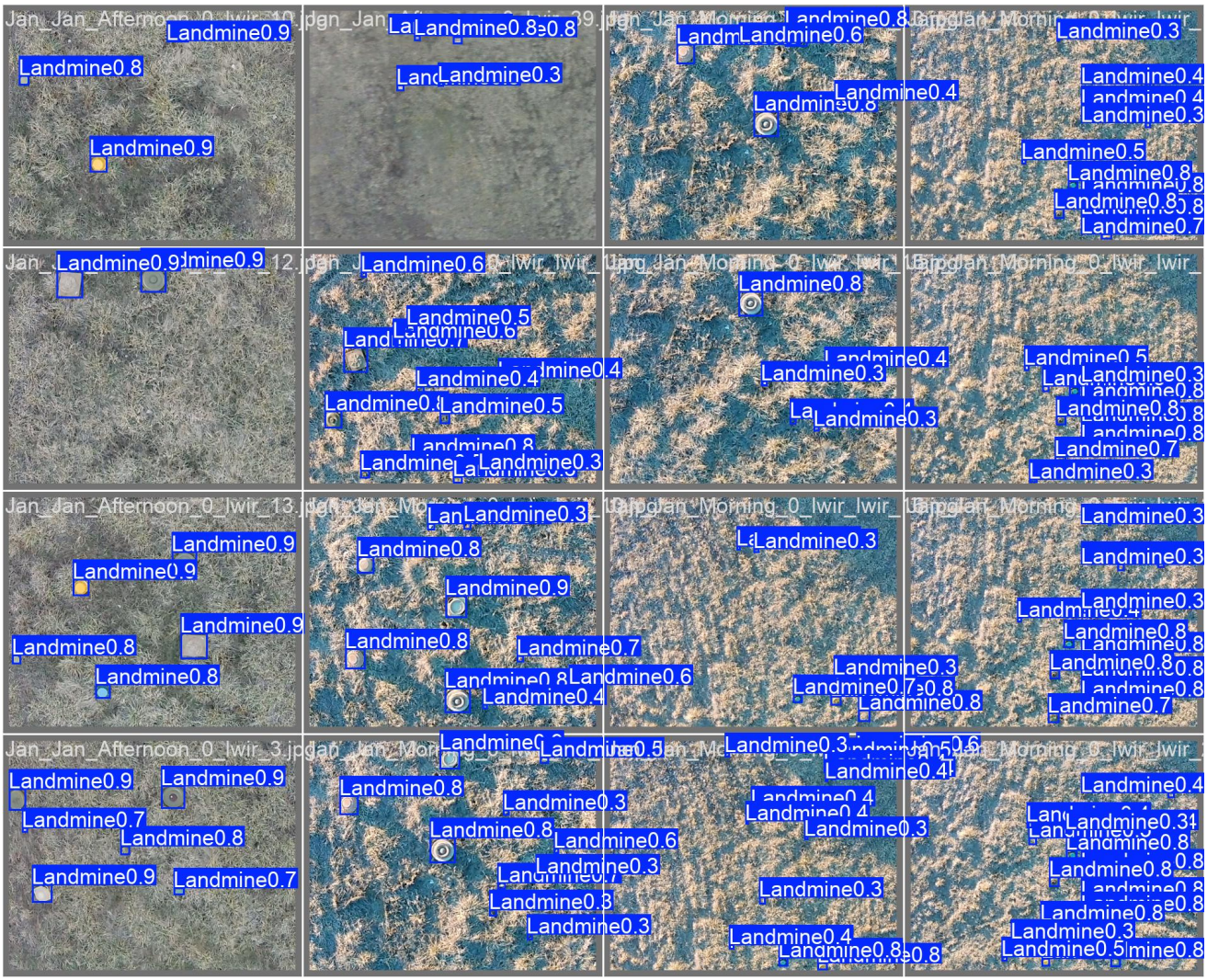




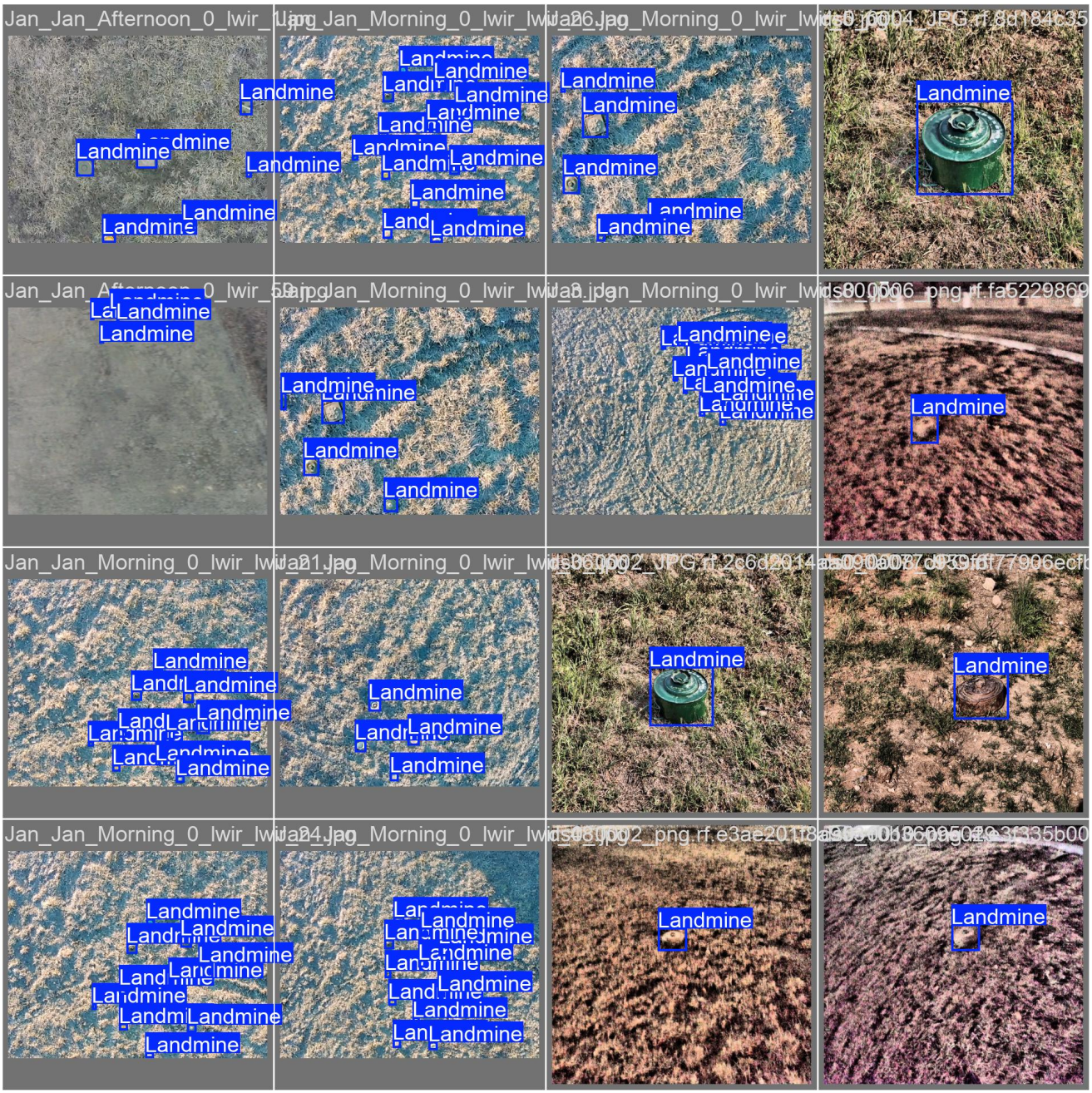
Validation Ground Truth (batch 0)



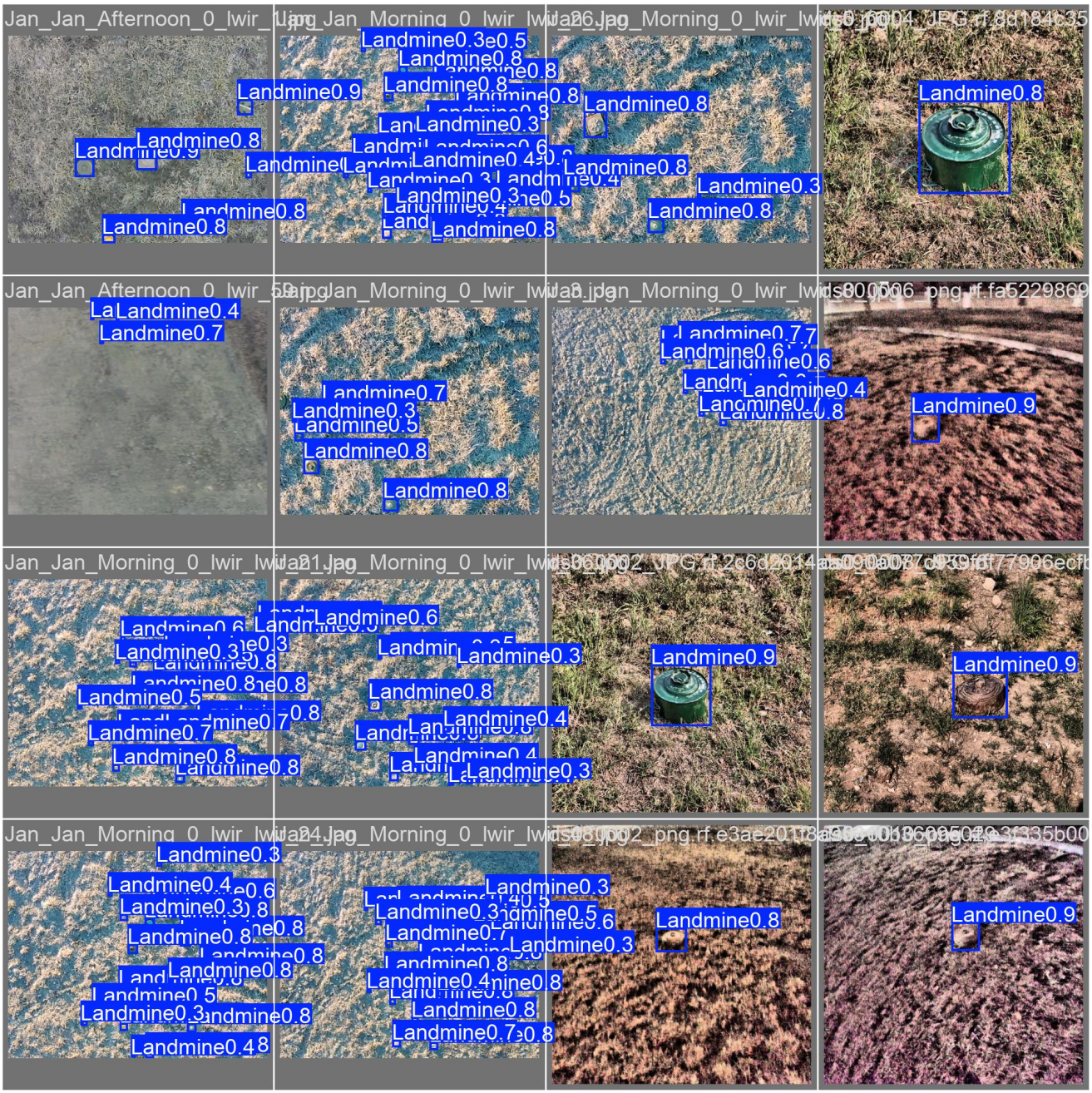
Validation Predictions (batch 0)



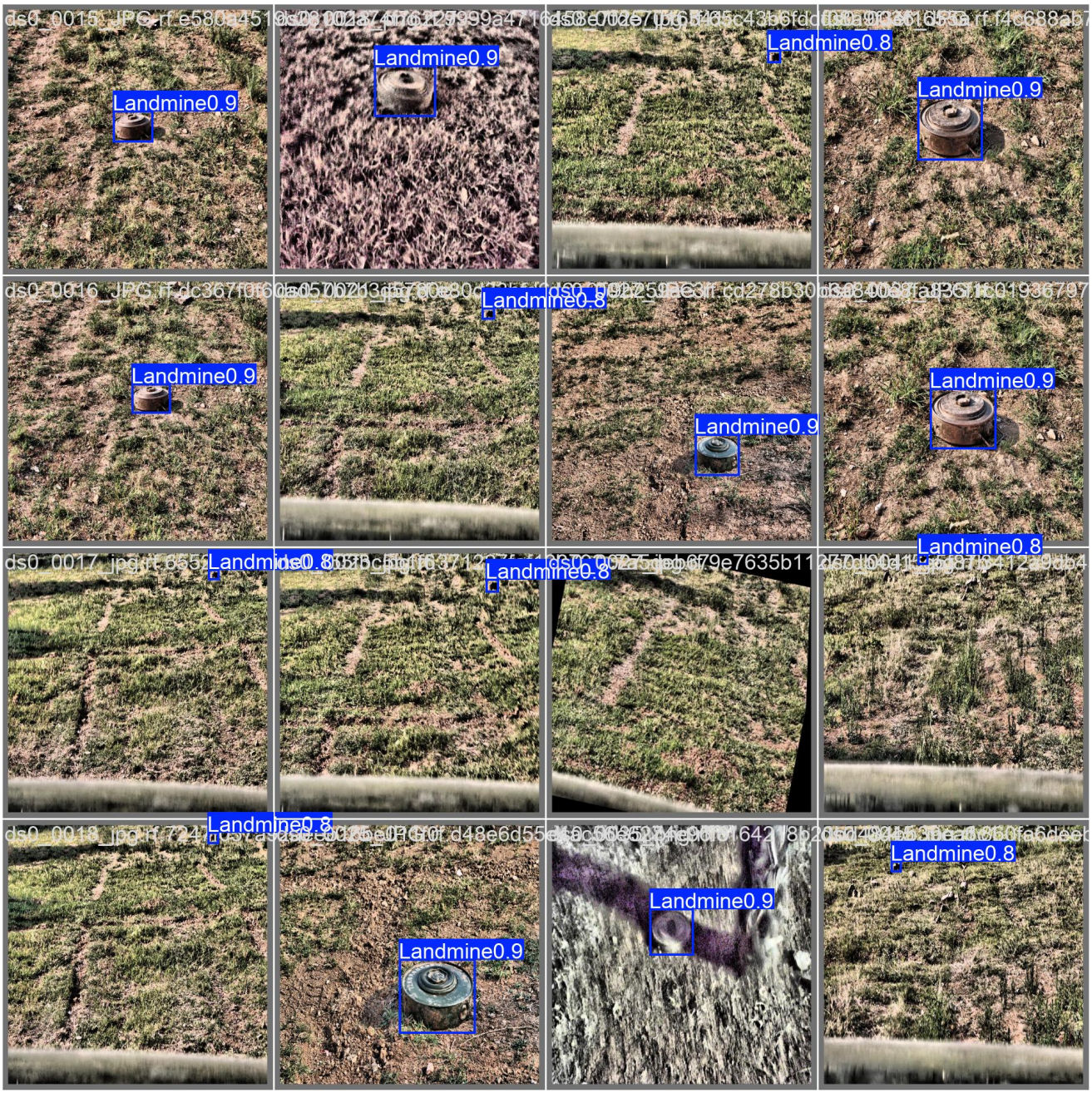
Test Ground Truth (batch 0) — seed 2



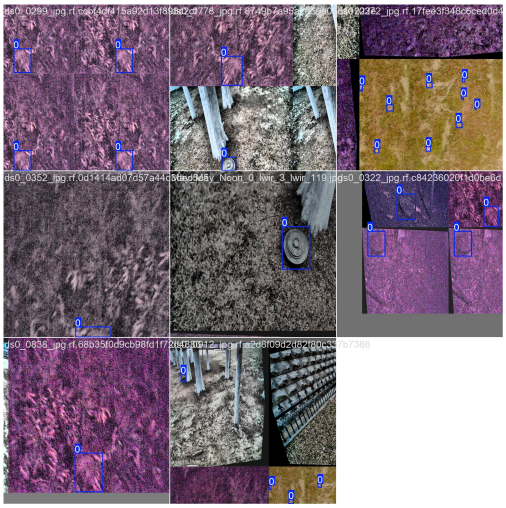
Test Predictions (batch 0) @ conf 0.21 — seed 2

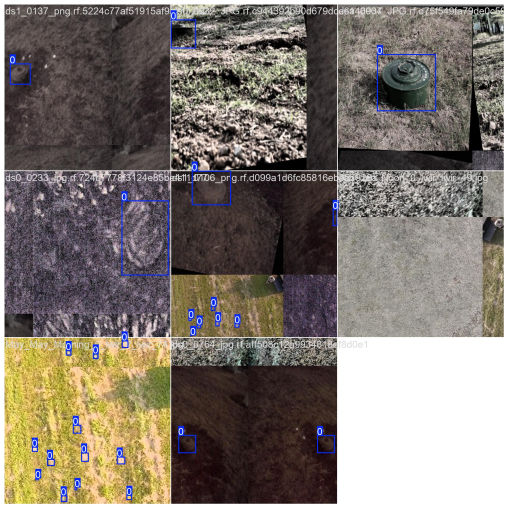


Test Predictions (batch 1) @ conf 0.21 — seed 2



Training Augmentation Sample





Official metrics come from paper_results/yolo/s_seed2/{val,test}.json (frozen protocol; OP conf chosen on val only). Test batch images above are from a visualization-only pass (make_test_vis.py); they do not change any reported number.